

Question Paper

Exam Date & Time: 24-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
FIFTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Natural Language Processing [CSE_3126]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Analyse how each stage of the NLP pipeline contributes to transforming raw text into meaningful representations, highlighting the interdependencies and possible challenges between stages with neat diagram. (4)
- A)
- Course Outcome: "CO1" / Cognitive Level: "Analysis"
- B) Apply the concept of morphology to analyse the structure of the following words: *unhappiness*, *restarted*, and *internationally* (4)
- Course Outcome: "CO1" / Cognitive Level: "Application"
- C) "Given two sentences - (4)
- (a) 'He studies hard every day.'
- (b) 'He studied hard yesterday.'
- Apply lemmatization and show how it helps in treating semantically similar words equally during model training."
- Course Outcome: "CO1" / Cognitive Level: "Application"
- 2) Estimate how well a Finite-State Transducer can generate English verb inflections (e.g., *talk* → *talking*, *talked*). Construct a small FST for these forms and compare its effectiveness to languages with richer morphology. (4)
- A)
- Course Outcome: "CO2" / Cognitive Level: "Evaluate"
- B) Evaluate the effectiveness of the Minimum Edit Distance algorithm for spelling-correction and string transformation by calculating the distance between PAPER → PEPPER using the operation costs (insertion = 1, deletion = 1, substitution = 2). Construct the complete distance matrix and perform backtracking to identify the optimal edit sequence. (4)
- Course Outcome: "CO2" / Cognitive Level: "Evaluate"
- C) Compare lexicon-based FSTs and lexicon-free FSTs, providing one advantage of each. (2)
- Course Outcome: "CO2" / Cognitive Level: "Understand"
- 3) Consider the sentence: (4)
- "The students read books."
- A) Write the probability expression: (4)
- (a) Without the Markov Assumption

(b) With the Bigram Assumption
Compare the complexity of both expressions.

Course Outcome: "CO3" / Cognitive Level: "Evaluate"

- B) Given the corpus: (4)
"students play games
students play football"
Apply Laplace Smoothing, for the bigram "Play Games".

Course Outcome: "CO3" / Cognitive Level: "Application"

- C) Explain how unsmoothed n-gram models often assign zero probability to unseen word sequences? (2)

Course Outcome: "CO3" / Cognitive Level: "Understand"

- 4) Distinguish between open class words and closed class words in English. Provide two examples for each type and discuss their major grammatical and semantic characteristics. (4)

- A) Course Outcome: "CO4" / Cognitive Level: "Analysis"

- B) Evaluate with the **Noisy Channel Model**, the misspelled word "**recieve**" may correspond to one of the following intended words: "**receive**," "**recipe**," or "**recite**." (4)

The model parameters are provided below:

| Candidate Word (w) | Language Model Probability P(w) | Error Model Probability P(x|w) |
Joint Probability P(w)×P(x|w) |

Word	P(word)	P(observe d word)
receive	0.004	0.2
recipe	0.001	0.02
recite	0.0003	0.01

Evaluate which word is the **most probable intended correction** for "recieve."

Course Outcome: "CO4" / Cognitive Level: "Evaluate"

- C) Explain four major English word classes used in POS tagging and illustrate each with a suitable example. (2)

Course Outcome: "CO4" / Cognitive Level: "Remember"

- 5) Evaluate the grammatical structure of the sentence "**The students submitted their assignments on time**" by constructing and labelling its dependency parse tree. Justify your choice of head word and explain how each dependency relation supports the overall syntactic structure. (4)

- A) Course Outcome: "CO5" / Cognitive Level: "Evaluate"

- B) Analyze how top-down parsing applies the given grammar rules to derive the sentence "The young boy saw the big dog in the park." (ART stands for article, ADJ for adjective, and PP for prepositional phrase.) (4)

S → NP VP

NP → ART N

NP → ART ADJ N

NP → NP PP

VP → V NP
 PP → P NP
 ART → the
 ADJ → young | big
 N → boy | dog | park
 V → saw
 P → in

Course Outcome: "CO5" / Cognitive Level: "Analysis"

- C) **Explain why the CKY (Cocke-Kasami-Younger) parsing algorithm requires the grammar to be in *Chomsky Normal Form (CNF)*.** Using the following grammar rules, demonstrate one step of the CKY dynamic programming table for the sentence "**a dog barks.**" (2)

S → NP VP
 NP → ART N
 VP → V
 ART → a
 N → dog
 V → barks

Course Outcome: "CO5" / Cognitive Level: "Understand"

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Event Name	Subject
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058-END SEMSTER-ODD-2025 CSE_3126:Natural Language Processing

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025 CSE_3126:Natural Language Processing

#	Level	No of Questions	Marks (%)
1	Remember	1	2(4%)
2	Understand	3	6(12%)
3	Application	3	10(20%)
4	Analysis	3	12(24%)
5	Create	5	20(40%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)